PRICING AND HEDGING OF RENEWABLE ENERGY QUANTO OPTIONS IN CARMA MODELS

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Abstract. This paper investigates the pricing and hedging of Quanto options in renewable energy markets, utilizing continuous-time autoregressive moving-average (CARMA) processes to model the dynamics of electricity spot prices and weather-related variables. Quanto options, which are European-style derivatives, are written on both the electricity spot or forward price and a weather or climate variable such as temperature, wind speed, or solar irradiance. These options provide renewable energy firms with the means to hedge against both price fluctuations and volumetric risk, enabling better risk management in the renewable energy sector. We demonstrate how CARMA models offer a flexible and powerful framework for accurately pricing these contracts while effectively addressing both the price and weather-dependent risk. By applying CARMA-driven models to joint price and weather dynamics, we extend the classical Ornstein-Uhlenbeck processes to capture more complex market behaviors, in particular allow for a more complex autocorrelation structure of the underyling price and weather components.

Keywords: Quanto Options, Renewable Energy Markets, CARMA, Volumetric Risk, Weather Derivative

1. Introduction

Continuous-time autoregressive moving average (CARMA) processes have gained significant attention in financial modeling due to their flexibility and ability to capture complex time series dynamics. By generalizing the classical discrete-time ARMA models to continuous time, CARMA processes offer a robust framework for modeling phenomena where continuous-time dynamics are essential, such as in the pricing and hedging of financial derivatives. These processes provide a rich structure for capturing various characteristics of time series data, including mean reversion, volatility clustering, and long-memory effects, which are prevalent in financial and energy markets.

CARMA processes have been widely applied across various fields, from engineering and meteorology to finance, particularly in modeling interest rates, stochastic volatility, and commodity prices [**? ? ?**]. In the context of energy markets, accurately modeling the dynamics of variables such as wind speed, solar irradiance, and temperature is crucial. These environmental variables exhibit complex behaviors that traditional models, such as Ornstein-Uhlenbeck (OU) processes, may not fully capture. For example, wind speed and solar irradiance have been successfully modeled using higher-order ARMA and CARMA processes to account for their temporal dependencies and autocovariance structures [].

In recent years, there has been growing interest in hybrid derivatives that combine price and volumetric risk, particularly in the energy sector, where weather variability can significantly impact both energy production and demand. Renewable energy quanto options are a prime example of such derivatives, as their payoffs

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depend on both the electricity spot price and a weather-related variable such as temperature. Previous work by Alfonsi and Vadillo [**?**] modeled day-ahead electricity prices and average daily temperatures using non-homogeneous Ornstein-Uhlenbeck processes, demonstrating the relevance of weather-driven models for energy derivatives. However, their use of OU processes, while tractable, may not capture the full complexity of the market behavior.

To address these limitations, we propose modeling both energy prices and weather-related variables such as temperature using higher-order CARMA processes. CARMA processes offer greater flexibility by allowing for more sophisticated autocovariance structures and the ability to capture complex market behaviors such as volatility clustering, jumps, and heavy tails, which are frequently observed in energy prices and weather data []. For instance, temperature variations, a key driver of energy demand, have been effectively modeled using CARMA processes, providing a better fit to data with long-range dependencies and mean-reversion []. Similarly, solar irradiance and wind speed, both crucial for renewable energy production, can be modeled using ARMA and CARMA models of appropriate order to capture their stochastic nature [].

1.1. **Background on CARMA Processes.** Continuous-time autoregressive moving average (CARMA) processes extend the classical discrete-time ARMA models to continuous time, offering a powerful tool for modeling continuous-time stochastic dynamics. They are defined as solutions to higher-order stochastic differential equations (SDEs) of the form:

$$\mathrm{D}^p X_t + a_1 \mathrm{D}^{p-1} X_t + \cdots + a_p X_t = b_0 \mathrm{D}^q L_t + b_1 \mathrm{D}^{q-1} L_t + \cdots + b_q L_t, \qquad (1)$$

where $\mathrm{D} = \dfrac{\mathrm{d}}{\mathrm{d}t}$ denotes differentiation, a_i and b_j are coefficients, and L_t is a Lévy process driving the system. This equation generalizes classical ARMA models to continuous time, capturing dynamics with rich temporal dependencies and flexible autocovariance structures.

In financial applications, CARMA processes are particularly valuable for modeling time series that exhibit mean reversion and short-memory behavior—common features in financial data such as asset returns, interest rates, and volatility. The Ornstein-Uhlenbeck process is a well-known example of a CARMA(1,0) process, widely used due to its tractability and mean-reverting properties. However, higher-order CARMA processes provide greater flexibility by allowing more complex autocorrelation structures, capable of capturing a wider range of dynamic behaviors observed in financial markets.

1.2. **Relevance to Energy Markets.** Energy markets are characterized by significant volatility and are influenced by various stochastic factors, including commodity prices and weather variables like temperature. Accurate modeling of these factors is crucial for pricing and hedging energy derivatives, such as energy quanto options, which are financial instruments whose payoffs depend on both the price of a commodity and a weather-related variable.

Traditional models often rely on Ornstein-Uhlenbeck processes to model these factors; however, the limitations of OU processes—such as their inability to capture complex temporal dependencies and higher-order autocorrelations—motivate the use of higher-order CARMA processes. By modeling both the commodity price and the temperature using higher-order CARMA processes driven by Lévy processes, we can capture the intricate dynamics and interactions between these variables more accurately.

In our framework, we consider correlated CARMA processes to model the joint behavior of the commodity price and temperature, capturing both individual dynamics and co-movement between these variables. This joint modeling is essential for accurately pricing energy quanto options and developing effective hedging strategies.

1.3. **Main Contributions.** In this paper, we propose a joint modeling framework for energy spot prices and temperature indices driven by correlated CARMA processes. This extension provides several advantages over traditional OU-based models, particularly in capturing the intricate behaviors observed in real market data. By introducing multivariate Lévy processes to drive these dynamics, we accommodate jumps, heavy tails, and skewness, which are often present in energy prices and other financial time series. Additionally, the CARMA framework allows for a parsimonious state-space representation, enabling efficient pricing and hedging of complex derivatives like energy quanto options.

Our contributions are as follows:

(1) **Extension of Energy Market Modeling**: We incorporate higher-order CARMA processes for both commodity prices and temperature, providing a more flexible and accurate framework compared to traditional OU models. This allows for capturing complex temporal dependencies and interactions between these variables.
(2) **Derivation of Pricing Formulas**: We derive analytical pricing formulas and efficient numerical methods for energy quanto options within this framework, accounting for complex covariance structures and potential jumps in the underlying processes.
(3) **Development of Hedging Strategies**: We compute the Greeks for the options and develop hedging strategies that allow practitioners to effectively manage risks associated with movements in both commodity prices and temperature.

1.4. **Organization of the Paper.** The remainder of this paper is organized as follows. In Section **??**, we provide a detailed overview of CARMA processes, including their mathematical properties and state-space representations. Section 2 introduces the modeling framework for energy markets using correlated CARMA processes and discusses the derivation of covariance structures and conditional expectations necessary for option pricing. In Section 3, we present the pricing methodology for energy quanto options, including the derivation of analytical expressions and considerations for numerical implementation. Section **??** discusses hedging strategies, including the computation of the Greeks and the construction of hedging portfolios. Finally, Section **??** concludes the paper and suggests directions for future research.

2. Coupled CARMA Models for Renewable Energy Markets

In this section, we introduce a coupled CARMA model to capture the dynamics of time-dependent variables in renewable energy markets. Specifically, we focus on the joint behavior of power prices and renewable energy sources such as temperature, wind speed, or solar irradiation, all of which can be effectively modeled by higher-order CARMA processes, as outlined in the introduction. Accurate modeling of these factors, particularly the covariance structure, is essential for pricing and hedging renewable energy quanto options, which depend on both energy prices and weather-related variables.

2.1. **Notation and Preliminaries.** Throughout this paper, we adopt the following notation. Let $\mathbb{N}$ denote the set of natural numbers $\{1, 2, 3, \ldots\}$, and let $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ represent the set of non-negative integers. For a complex number $z = a + \mathrm{i}b \in \mathbb{C}$, its real part is denoted by $\Re(z) = a$ and its imaginary part by $\Im(z) = b$. For $d \in \mathbb{N}$, we let $\mathbb{R}^d$ denote the d-dimensional Euclidean space equipped with the standard inner product $\langle \cdot, \cdot \rangle$, while $\{e_1, e_2, \ldots, e_d\}$ represents the standard basis vectors in $\mathbb{R}^d$. We also denote by $\mathbb{R}^{n\times m}$ the space of real $n \times m$ matrices, and by $\mathbb{R}^n$ the space of real n-dimensional column vectors. The Euclidean norm of a vector $\mathbf{x} \in \mathbb{R}^d$ is defined as $\|\mathbf{x}\| = \sqrt{\sum_{i=1}^d x_i^2}$.

2.1.1. *Lévy Processes.* Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions of completeness and right-continuity. Let $\{L_t\}_{t\geq 0}$ be a real-valued Lévy process defined on this probability space. Recall that a Lévy process is a stochastic process with stationary and independent increments, càdlàg paths, and $L_0 = 0$ almost surely.

The characteristic function of L_t is given by

$$\mathbb{E}\left[e^{\mathrm{i}\theta L_t}\right] = e^{t\psi_L(\theta)},$$

where $\psi_L(\theta)$ is the characteristic exponent, given by the Lévy-Khintchine formula:

$$\psi_L(\theta) = \mathrm{i}\theta\gamma - \frac{1}{2}\theta^2\sigma^2 + \int_{\mathbb{R}\setminus\{0\}} \left(e^{\mathrm{i}\theta x} - 1 - \mathrm{i}\theta x \mathbb{1}_{\{|x|\leq 1\}}\right) \nu(\mathrm{d}x), \tag{2}$$

for $\theta \in \mathbb{R}$. Here, $\gamma \in \mathbb{R}$ is the drift parameter, $\sigma^2 \geq 0$ is the variance of the Gaussian component, and ν is a measure on $\mathbb{R} \setminus \{0\}$ satisfying

$$\int_{\mathbb{R}\setminus\{0\}} (1 \wedge x^2)\, \nu(\mathrm{d}x) < \infty.$$

The measure ν is called the Lévy measure of the process L.

We say that a Lévy process L is *integrable* if $\mathbb{E}[|L_t|] < \infty$ for all $t \geq 0$, and *square-integrable* if $\mathbb{E}[L_t^2] < \infty$ for all $t \geq 0$. For a square-integrable Lévy process L, the mean of L_1 is denoted by μ_L and is given by

$$\mu_L = \gamma + \int_{|x|>1} x\, \nu(\mathrm{d}x).$$

Moreover, the variance of L_1 is given by

$$\mathrm{Var}(L_1) = \sigma^2 + \int_{\mathbb{R}} x^2\, \nu(\mathrm{d}x).$$

For any Lévy process $\{L_t^{(1)}\}_{t\geq 0}$ defined on the positive real line $\mathbb{R}_+$, we can define a *two-sided Lévy process* $\{L_t\}_{t\in\mathbb{R}}$ by choosing a second independent and identically

distributed Lévy process $\{L_t^{(2)}\}_{t\geq 0}$ and setting

$$L_t = \begin{cases} L_t^{(1)}, & \text{if } t \geq 0, \\ -L_{-t-}^{(2)}, & \text{if } t < 0, \end{cases} \tag{3}$$

where $L_{t-}^{(2)} = \lim_{s\nearrow t} L_s^{(2)}$ for all $t \geq 0$. This construction allows us to define processes over the entire real line.

Example 2.1 (Normal Inverse Gaussian Process)**.** The Normal Inverse Gaussian (NIG) process is a Lévy process defined by the characteristic function:

$$\mathbb{E}\left[e^{\mathrm{i}\theta L_t}\right] = \exp\left(\mathrm{i}\theta\mu t - \delta t\left(\sqrt{\alpha^2-\beta^2} - \sqrt{\alpha^2-(\beta+\mathrm{i}\theta)^2}\right)\right),$$

where the parameters $\alpha > 0$, $|\beta| < \alpha$, $\mu \in \mathbb{R}$, and $\delta > 0$ control the shape of the distribution. Here α governs the steepness of the distribution's tails; β controls the asymmetry or skewness of the distribution; δ is a scaling parameter and μ represents the drift.

Example 2.2 (Alpha-Stable Lévy process)**.** The alpha-stable Lévy process is a Lévy process with the following characteristic function:

$$\mathbb{E}\left[e^{\mathrm{i}\theta L_t}\right] = \exp\left(t\left(\mathrm{i}\theta\gamma - \sigma^\alpha|\theta|^\alpha(1-\mathrm{i}\beta\,\mathrm{sign}(\theta)\tan\left(\frac{\pi\alpha}{2}\right))\right)\right),$$

where the parameters satisfy $0 < \alpha \leq 2$, $|\beta| \leq 1$, $\sigma > 0$, and $\gamma \in \mathbb{R}$. Here α is the stability parameter, controlling the heaviness of the tails (with smaller values of α corresponding to heavier tails); β is the skewness parameter, determining the asymmetry of the distribution; σ is the scale parameter, controlling the dispersion of the process and γ is the drift parameter, governing the location of the distribution.

Alpha-stable Lévy processes are known for their ability to model distributions with infinite variance (for $\alpha < 2$), which makes them suitable for capturing extreme market events, such as price spikes or significant drops, that may occur in energy markets. These processes are also characterized by heavy tails, which are often observed in financial and energy data.

In the context of energy markets, alpha-stable Lévy processes can be used to model the stochastic dynamics of prices or demand, allowing for the possibility of large, infrequent changes in market conditions. This feature makes alpha-stable processes particularly useful for modeling renewable energy sources where variability is driven by extreme weather conditions (e.g., wind gusts or sudden changes in solar irradiance).

2.1.2. *Lévy-Driven CARMA(p,q) Processes.* To model the joint dynamics of commodity prices and temperature, we employ coupled CARMA processes driven by real-valued Lévy processes. This approach captures both the temporal dependencies within each variable and the cross-dependencies between them, which are essential for accurately pricing and hedging renewable energy quanto options.

Definition 2.3. *A* Lévy-driven CARMA(p,q) process*, with* $0 \leq q < p$*, is defined as* $Y(t) = C^\top \mathbf{X}(t)$ *(output equation), where* $\mathbf{X}(t) \in \mathbb{R}^p$ *is the solution of the following stochastic differential equation (state equation):*

$$\mathrm{d}\mathbf{X}(t) = A\mathbf{X}(t)\,\mathrm{d}t + B\,\mathrm{d}L(t), \tag{4}$$

with

$$A = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_p & -a_{p-1} & -a_{p-2} & \cdots & -a_1 \end{pmatrix}, \tag{5}$$

$$B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}, \quad C = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_p \end{pmatrix}, \tag{6}$$

where $a_i, b_j \in \mathbb{R}$ are real coefficients with $b_j \neq 0$ for $1 \leq j \leq q+1$ and $b_j = 0$ for $q+2 \leq j \leq p$. Here, $\{L(t)\}_{t\in\mathbb{R}}$ is a real-valued Lévy process. The matrix A is a $p \times p$ companion matrix containing the coefficients of the autoregressive part, which relate to the stationarity and mean-reversion properties of the process.

Remark 2.4. *The structure of the matrices and vectors in the CARMA model ensures that the process $Y(t)$ satisfies the following higher-order stochastic differential equation:*

$$D^pY(t) + a_1D^{p-1}Y(t) + \cdots + a_pY(t) = b_0D^qL(t) + b_1D^{q-1}L(t) + \cdots + b_qL(t), \tag{7}$$

where $D = \dfrac{\mathrm{d}}{\mathrm{d}t}$ denotes differentiation. This equation highlights the autoregressive (AR) and moving average (MA) components of the CARMA process.

Proposition 2.5. *A solution of the state equation is a p-dimensional Ornstein-Uhlenbeck process given by the variation-of-constants formula:*

$$\mathbf{X}_t = e^{(t-s)A}\mathbf{X}_s + \int_s^t e^{(t-u)A}\mathbf{e}_p \,\mathrm{d}L_u, \quad 0 \leq s < t. \tag{8}$$

Remark 2.6.

- *By setting $p = 1$, we have $A = -\kappa = -a_1$. This leads us to the Ornstein-Uhlenbeck (OU) process, which has been widely applied in finance, particularly in option pricing. Therefore, CARMA processes are higher-order generalizations of OU processes:*

$$\mathrm{d}X(t) = -\kappa X(t)\,\mathrm{d}t + \mathrm{d}L(t).$$

- *A CARMA process can also be represented as the solution of the following stochastic differential equation (SDE):*

$$a(D)Y(t) = b(D)DL(t),$$

where D denotes differentiation with respect to t, and $a(\cdot)$ and $b(\cdot)$ are polynomials given by

$$a(z) = z^p + a_1z^{p-1} + \cdots + a_p,$$
$$b(z) = b_0 + b_1z + \cdots + b_qz^q.$$

- *We will assume that $\mathbf{b} = \mathbf{e}_1$, i.e., $b_1 = 1$ and $b_j = 0$ for $j > 1$. Furthermore, by setting $q = 0$, the* **CARMA**(p, q) *process becomes a* **CAR**(p) *process.*

Therefore, we can write the **CAR**(p) *process as:*

$$\begin{cases} \mathrm{d}X_t^{(i)} = X_t^{(i+1)}\,\mathrm{d}t, & i = 1, 2, \ldots, p-1, \\ \mathrm{d}X_t^{(p)} = -\sum_{k=1}^{p} a_k X_t^{(k)}\,\mathrm{d}t + \mathrm{d}L_t, \\ Y(t) = X_t^{(1)}. \end{cases}$$

2.2. **Modelling the Spot and Weather with Coupled CARMA Processes.** To accurately capture the combined dynamics of daily day-ahead electricity log spot prices $\{X_t\}_{t\geq 0}$ and daily average temperatures $\{T_t\}_{t\geq 0}$, we propose a coupled stochastic model based on higher-order continuous-time autoregressive (CAR(p)) processes. This approach extends the standard Ornstein-Uhlenbeck process (CAR(1)) to CAR(p) processes, allowing for more complex temporal dependencies and improved modeling of the observed data.

Our model builds upon the Electricity Temperature Model (ETM) described in [**?**], replacing the CAR(1) processes with CAR(p) processes to capture long-memory effects and intricate dynamics inherent in energy markets. By considering higher-order processes, we aim to provide a more flexible and accurate representation of the underlying stochastic behaviors.

Both the electricity prices and temperatures are assumed to be *deseasonalized*, meaning that deterministic seasonal patterns have been removed from the data. This preprocessing step allows us to focus on modeling the stochastic fluctuations around these seasonal trends.

2.2.1. *Temperature Process.* We model the deseasonalized temperature process using the following state-space representation:

$$\begin{cases} \mathrm{d}\mathbf{Z}_t^T = A_T \mathbf{Z}_t^T\,\mathrm{d}t + \mathbf{e}_p \sigma_T\,\mathrm{d}W_t^T, \\ T_t = \mathbf{b}_T^\top \mathbf{Z}_t^T, \end{cases} \tag{9}$$

where:

- $\mathbf{Z}_t^T \in \mathbb{R}^p$ is the state vector of the temperature process at time t.
- $A_T \in \mathbb{R}^{p\times p}$ is the state-transition matrix governing the dynamics of the temperature process. It is typically structured to reflect the coefficients of an autoregressive polynomial of order p.
- $\sigma_T > 0$ is the volatility parameter of the temperature process.
- $\mathbf{e}_p \in \mathbb{R}^p$ is the p-th standard basis vector, defined as $\mathbf{e}_p = (0, 0, \ldots, 0, 1)^\top$.
- W_t^T is a standard Brownian motion representing the stochastic noise driving the temperature process.
- $\mathbf{b}_T \in \mathbb{R}^p$ is the output vector mapping the state vector $\mathbf{Z}_t^T$ to the observable temperature T_t.

The dynamics of $\mathbf{Z}_t^T$ are influenced by both the deterministic evolution dictated by A_T and the stochastic fluctuations introduced by the Brownian motion W_t^T. The observable temperature T_t is obtained as a linear combination of the state variables, allowing for flexibility in capturing the temperature dynamics.

2.2.2. *Electricity Log Spot Price Process.* The deseasonalized electricity log spot price process is modeled using a CAR(p) process that incorporates dependence on the temperature process and accounts for jumps through a Lévy process:

$$\begin{cases} \mathrm{d}\mathbf{Z}_t^X = A_X \mathbf{Z}_t^X \ \mathrm{d}t + \lambda \sigma_T \mathbf{e}_p \ \mathrm{d}W_t^T + \mathbf{e}_p \ \mathrm{d}L_t^X, \\ X_t = \mathbf{b}_X^\top \mathbf{Z}_t^X, \end{cases} \tag{10}$$

where:

- $\mathbf{Z}_t^X \in \mathbb{R}^p$ is the state vector of the electricity price process at time t.
- $A_X \in \mathbb{R}^{p \times p}$ is the state-transition matrix for the electricity price process.
- $\lambda \in \mathbb{R}$ is a coupling parameter that introduces dependence between the temperature and electricity price processes.
- L_t^X is a real-valued Lévy process with Normal Inverse Gaussian (NIG) distribution parameters $(\alpha^X, \beta^X, \delta^X, \mu^X)$, capturing the jump behavior commonly observed in electricity prices.
- $\mathbf{b}_X \in \mathbb{R}^p$ is the output vector for the electricity price process.

The term $\lambda \sigma_T \mathbf{e}_p \ \mathrm{d}W_t^T$ introduces dependence between the electricity price and temperature processes through the shared Brownian motion W_t^T. The coupling parameter λ controls the strength and direction of this dependence, allowing us to model positive or negative correlations between temperature and electricity prices. The inclusion of the Lévy process L_t^X accounts for sudden spikes and heavy-tailed behavior in electricity prices due to market shocks or unexpected events.

2.2.3. *Integrated Form of the Model.* By applying the variation of constants formula to the state equations, we derive the integrated forms of the state processes for all $s < t$:

$$\begin{cases} \mathbf{Z}_t^T = e^{A_T(t-s)} \mathbf{Z}_s^T + \sigma_T \displaystyle\int_s^t e^{A_T(t-u)} \mathbf{e}_p \ \mathrm{d}W_u^T, \\ \mathbf{Z}_t^X = e^{A_X(t-s)} \mathbf{Z}_s^X + \lambda \sigma_T \displaystyle\int_s^t e^{A_X(t-u)} \mathbf{e}_p \ \mathrm{d}W_u^T + \int_s^t e^{A_X(t-u)} \mathbf{e}_p \ \mathrm{d}L_u^X. \end{cases} \tag{11}$$

In these expressions:

- $e^{A_T(t-s)}$ and $e^{A_X(t-s)}$ represent the deterministic evolution of the state vectors due to the autoregressive components.
- The stochastic integrals capture the cumulative effect of the stochastic inputs over the interval $[s, t]$.
- The shared Brownian motion W_t^T in both integrals introduces correlation between the temperature and electricity price processes.
- The integral with respect to the Lévy process L_t^X accounts for the jump contributions to the electricity price dynamics, reflecting sudden market changes.

Remark 2.7. *The coupling parameter λ plays a crucial role in modeling the correlation between temperature and electricity prices. A positive λ indicates that increases in temperature are associated with increases in electricity prices, which may occur during periods of high demand due to heating or cooling needs. Conversely, a negative λ suggests an inverse relationship.*

2.3. **Covariance of Residuals.** Understanding the covariance between the residuals of the temperature and electricity price processes is crucial for accurate modeling and risk management in energy markets. In this subsection, we derive expressions for the covariance of these residuals within our coupled CAR(p) framework.

Proposition 2.8. *Let $T_t = \mathbf{b}_T^\top \mathbf{Z}_t^T$ and $X_t = \mathbf{b}_X^\top \mathbf{Z}_t^X$ represent the temperature and electricity price processes, respectively. Then, the covariance between T_t and X_t is given by*

$$\mathrm{Cov}(T_t, X_t) = \mathbf{b}_T^\top \mathrm{Cov}\left(\mathbf{Z}_t^T, \mathbf{Z}_t^X\right) \mathbf{b}_X. \tag{12}$$

Proof. Since T_t and X_t are linear functions of their respective state vectors, we have

$$\begin{aligned}\mathrm{Cov}(T_t, X_t) &= \mathrm{Cov}\left(\mathbf{b}_T^\top \mathbf{Z}_t^T,\, \mathbf{b}_X^\top \mathbf{Z}_t^X\right)\\ &= \mathbf{b}_T^\top \mathrm{Cov}\left(\mathbf{Z}_t^T, \mathbf{Z}_t^X\right) \mathbf{b}_X,\end{aligned}$$

where we used the linearity of the covariance operator. □

Remark 2.9. *This result highlights that the covariance between the observable processes T_t and X_t depends on the covariance between their state vectors $\mathbf{Z}_t^T$ and $\mathbf{Z}_t^X$ and the output vectors $\mathbf{b}_T$ and $\mathbf{b}_X$. Understanding this relationship is essential for capturing the co-movement between temperature and electricity prices.*

Next, we derive an explicit expression for the covariance of the residuals of the state vectors over a time increment Δ.

Proposition 2.10 (Covariance of Residuals)**.** *The covariance of the residuals of the state vectors over the interval $[t, t+\Delta]$ is given by*

$$\mathrm{Cov}\left(\mathbf{R}_t^T,\, \mathbf{R}_t^X\right) = \lambda \sigma_T^2\, e^{A_T \Delta} \hat{\mathcal{D}}_{TX}^{-1} \left(e^{\hat{\mathcal{D}}_{TX}\Delta} - \mathbf{I}\right) \mathbf{e}_p \mathbf{e}_p^\top e^{A_X^\top \Delta}, \tag{13}$$

where

$$\mathbf{R}_t^T = \mathbf{Z}_{t+\Delta}^T - e^{A_T\Delta}\mathbf{Z}_t^T, \tag{14}$$

$$\mathbf{R}_t^X = \mathbf{Z}_{t+\Delta}^X - e^{A_X\Delta}\mathbf{Z}_t^X, \tag{15}$$

and the operator $\hat{\mathcal{D}}_{TX} : \mathbb{R}^{p\times p} \to \mathbb{R}^{p\times p}$ is defined by

$$\hat{\mathcal{D}}_{TX}(M) = A_T M + M A_X^\top, \quad M \in \mathbb{R}^{p\times p}. \tag{16}$$

Proof. From the integrated forms of the state equations (11), we have

$$\begin{aligned}\mathbf{R}_t^T &= \sigma_T \int_t^{t+\Delta} e^{A_T(t+\Delta-u)}\mathbf{e}_p \,\mathrm{d}W_u^T,\\ \mathbf{R}_t^X &= \lambda\sigma_T \int_t^{t+\Delta} e^{A_X(t+\Delta-u)}\mathbf{e}_p \,\mathrm{d}W_u^T + \int_t^{t+\Delta} e^{A_X(t+\Delta-u)}\mathbf{e}_p \,\mathrm{d}L_u^X.\end{aligned}$$

Since W_t^T and L_t^X are independent processes, the covariance between $\mathbf{R}_t^T$ and the part of $\mathbf{R}_t^X$ involving $\mathrm{d}L_u^X$ is zero. Therefore, we have

$$\begin{aligned}\mathrm{Cov}\left(\mathbf{R}_t^T,\, \mathbf{R}_t^X\right) &= \mathrm{Cov}\left(\mathbf{R}_t^T,\, \lambda\sigma_T \int_t^{t+\Delta} e^{A_X(t+\Delta-u)}\mathbf{e}_p \,\mathrm{d}W_u^T\right)\\ &= \lambda\sigma_T^2 \int_t^{t+\Delta} e^{A_T(t+\Delta-u)}\mathbf{e}_p\mathbf{e}_p^\top e^{A_X^\top(t+\Delta-u)} \,\mathrm{d}u.\end{aligned}$$

Letting $\tau = t+\Delta-u$, the limits of integration become $\tau \in [0,\Delta]$, and we have

$$\mathrm{Cov}\left(\mathbf{R}_t^T,\, \mathbf{R}_t^X\right) = \lambda\sigma_T^2 \int_0^\Delta e^{A_T\tau}\mathbf{e}_p\mathbf{e}_p^\top e^{A_X^\top\tau} \,\mathrm{d}\tau.$$

This integral can be expressed using the operator $\hat{\mathcal{D}}_{TX}$ by noting that

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(e^{A_T\tau} M e^{A_X^\top\tau}\right) = e^{A_T\tau}\hat{\mathcal{D}}_{TX}(M) e^{A_X^\top\tau}.$$

Integrating both sides from 0 to Δ, we obtain

$$\int_0^\Delta e^{A_T\tau}\hat{\mathcal{D}}_{TX}(M)e^{A_X^\top\tau}\,\mathrm{d}\tau = e^{A_T\Delta}Me^{A_X^\top\Delta} - M.$$

Therefore,

$$\int_0^\Delta e^{A_T\tau}Me^{A_X^\top\tau}\,\mathrm{d}\tau = \hat{\mathcal{D}}_{TX}^{-1}\left(e^{A_T\Delta}Me^{A_X^\top\Delta} - M\right).$$

Applying this result with $M = \mathbf{e}_p\mathbf{e}_p^\top$, we have

$$\begin{aligned}\mathrm{Cov}\left(\mathbf{R}_t^T, \mathbf{R}_t^X\right) &= \lambda\sigma_T^2\hat{\mathcal{D}}_{TX}^{-1}\left(e^{A_T\Delta}\mathbf{e}_p\mathbf{e}_p^\top e^{A_X^\top\Delta} - \mathbf{e}_p\mathbf{e}_p^\top\right)\\ &= \lambda\sigma_T^2 e^{A_T\Delta}\hat{\mathcal{D}}_{TX}^{-1}\left(e^{\hat{\mathcal{D}}_{TX}\Delta} - \mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\top e^{A_X^\top\Delta},\end{aligned}$$

where we used the fact that $e^{A_T\Delta}\mathbf{e}_p\mathbf{e}_p^\top e^{A_X^\top\Delta} = e^{\hat{\mathcal{D}}_{TX}\Delta}\mathbf{e}_p\mathbf{e}_p^\top$. □

Remark 2.11. *The operator $\hat{\mathcal{D}}_{TX}$ encapsulates the combined dynamics of the temperature and electricity price state processes. The expression for the covariance of the residuals involves the inverse of $\hat{\mathcal{D}}_{TX}$, highlighting the complexity introduced by the coupling parameter λ and the matrices A_T and A_X.*

Next, we provide a lemma that is useful for calculating expectations involving the integral of an exponential matrix.

Lemma 2.12. *Let $A \in \mathbb{R}^{p\times p}$ be an invertible matrix, Y_t be a stochastic process with $\mathbb{E}[Y_t] = \mu_Y t$, and define the process*

$$\mathbf{X}_t = \int_0^t e^{A(t-s)}\mathbf{e}_p\,\mathrm{d}Y_s. \tag{17}$$

Then the expectation of $\mathbf{X}_t$ is

$$\mathbb{E}[\mathbf{X}_t] = \mu_Y A^{-1}\left(e^{At} - \mathbf{I}\right)\mathbf{e}_p. \tag{18}$$

Proof. Since $\mathbb{E}[\mathrm{d}Y_s] = \mu_Y\,\mathrm{d}s$, we have

$$\begin{aligned}\mathbb{E}[\mathbf{X}_t] &= \int_0^t e^{A(t-s)}\mathbf{e}_p\,\mathbb{E}[\mathrm{d}Y_s]\\ &= \mu_Y\int_0^t e^{A(t-s)}\mathbf{e}_p\,\mathrm{d}s.\end{aligned}$$

Making the substitution $\tau = t - s$, so $s = t - \tau$, and $\mathrm{d}s = -\,\mathrm{d}\tau$, we obtain

$$\begin{aligned}\mathbb{E}[\mathbf{X}_t] &= \mu_Y\int_0^t e^{A\tau}\mathbf{e}_p\,\mathrm{d}\tau\\ &= \mu_Y\left(\int_0^t e^{A\tau}\,\mathrm{d}\tau\right)\mathbf{e}_p\\ &= \mu_Y\left(A^{-1}\left(e^{At} - \mathbf{I}\right)\right)\mathbf{e}_p,\end{aligned}$$

where we used the fact that $\int_0^t e^{A\tau}\,\mathrm{d}\tau = A^{-1}(e^{At} - \mathbf{I})$ due to the invertibility of A. □

Remark 2.13. *The invertibility of A is crucial for the expression of $\mathbb{E}[\mathbf{X}_t]$. In a* **CAR**(p) *process, the matrix A is a companion matrix whose determinant is $\det(A) = (-1)^p a_p$, where $a_p \neq 0$ is the leading coefficient of the autoregressive polynomial. Therefore, $\det(A) \neq 0$ and A is invertible.*

2.3.1. *Implications for Modeling.* The derived expressions for the covariance of residuals are instrumental in understanding the joint behavior of the temperature and electricity price processes. They provide insights into how the coupling parameter λ and the system matrices A_T and A_X influence the covariance structure.

In practical applications, these results can be used for parameter estimation and model calibration. By matching the theoretical covariances with empirical estimates obtained from historical data, we can infer the values of λ, σ_T, and the entries of A_T and A_X.

Moreover, the covariance expressions are essential for risk management and derivative pricing. For instance, in the valuation of energy quanto options, understanding the covariance between temperature and electricity prices allows for more accurate modeling of the payoff distributions and hedging strategies.

2.3.2. *Additional Details on Covariance Calculation.* It's important to note that the covariance between the residuals arises solely due to the shared Brownian motion W_t^T and the coupling parameter λ. The Lévy process L_t^X, being independent of W_t^T, does not contribute to this covariance.

The operator $\hat{\mathcal{D}}_{TX}$ plays a central role in encapsulating the interaction between the two state processes. Its inverse appears in the covariance expression, reflecting the intertwined effects of the state dynamics governed by A_T and A_X.

3. Pricing of Energy Quanto Options in the Coupled CARMA Model

In this section, we develop a pricing framework for energy quanto options within the coupled CARMA model introduced earlier. We consider European-style options and derive the option pricing formulas by taking into account the joint dynamics of electricity prices and temperature.

3.1. **Option Payoff Structure.** Consider a European call option with maturity T written on the electricity spot price process $\{S_t\}_{t\geq 0}$, where the payoff also depends on the average temperature process $\{T_t\}_{t\geq 0}$. The payoff at maturity T is defined as

$$\text{Payoff} = \max\left(S_T - K,\, 0\right) \times f(T_T), \tag{19}$$

where

- K is the strike price.
- $f(T_T)$ is a function capturing the temperature dependence, for example, $f(T_T) = \exp(-\eta T_T)$ for some parameter $\eta \in \mathbb{R}$.

This payoff structure reflects a scenario where the option holder benefits from both favorable movements in the electricity price and certain temperature conditions, which is relevant for managing risks associated with weather-dependent energy production.

3.2. **Risk-Neutral Valuation Framework.** To price the option, we compute the expected discounted payoff under the risk-neutral measure $\mathbb{Q}$. The option price at time $t = 0$ is given by

$$V_0 = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\left[\max\left(S_T - K,\, 0\right) \times f(T_T)\,\middle|\,\mathcal{F}_0\right], \tag{20}$$

where

- V_0 is the option price at time $t = 0$.
- r is the risk-free interest rate.
- $\mathcal{F}_0$ represents the information available at time $t = 0$.

However, since temperature is not a tradable asset, we cannot hedge temperature risk perfectly, and thus, the standard risk-neutral valuation may not be directly applicable for the temperature component. One common approach is to assume that temperature risk is not priced, meaning that the market price of temperature risk is zero. This implies that we can work under the real-world measure $\mathbb{P}$ for the temperature process while using the risk-neutral measure $\mathbb{Q}$ for the tradable electricity price process.

3.3. **Dynamics Under the Risk-Neutral Measure.** Under the risk-neutral measure $\mathbb{Q}$, the dynamics of the electricity price process change to reflect the absence of arbitrage and the inclusion of risk premiums. Assuming that temperature risk is not priced, the dynamics of the temperature process remain the same under both $\mathbb{P}$ and $\mathbb{Q}$. The coupled CARMA model under $\mathbb{Q}$ is then given by

$$\begin{cases} \mathrm{d}\mathbf{Z}_t^T = A_T\mathbf{Z}_t^T\,\mathrm{d}t + \mathbf{e}_p\sigma_T\,\mathrm{d}W_t^T, \\ \mathrm{d}\mathbf{Z}_t^S = \left(A_S - \mathbf{e}_p\gamma_S\right)\mathbf{Z}_t^S\,\mathrm{d}t + \lambda\sigma_T\mathbf{e}_p\,\mathrm{d}W_t^T + \mathbf{e}_p\,\mathrm{d}L_t^{S,\mathbb{Q}}, \end{cases} \tag{21}$$

where

- $\mathbf{Z}_t^T \in \mathbb{R}^p$ is the state vector of the temperature process.
- $\mathbf{Z}_t^S \in \mathbb{R}^p$ is the state vector of the electricity price process.
- A_T, $A_S \in \mathbb{R}^{p\times p}$ are the state-transition matrices for temperature and electricity prices, respectively.

- $\sigma_T > 0$ is the volatility parameter for the temperature process.
- $\lambda \in \mathbb{R}$ is the coupling parameter between the two processes.
- W_t^T is a standard Brownian motion under $\mathbb{Q}$, driving the temperature process.
- $L_t^{S,\mathbb{Q}}$ is a Lévy process under $\mathbb{Q}$, representing the jump component in the electricity price dynamics.
- $\gamma_S \in \mathbb{R}$ is the market price of risk for the electricity price process, adjusting its drift under $\mathbb{Q}$.

Note that the temperature process remains unaffected by the change of measure, as we assume that temperature risk is not priced. The electricity price process, being a tradable asset, adjusts its drift through the risk premium parameter γ_S under the risk-neutral measure.

3.4. **General Payoff Structures.** In some cases, energy quanto options may have payoff functions involving sums over multiple time periods or combining put and call features. For example, consider a quanto option with payoff at time t defined as

$$\text{Payoff} = \sum_{t=t_1}^{t_2} f_S(S_t) \times f_T(T_t), \tag{22}$$

where f_S and f_T are functions such as

$$f_S(S_t) = \left(S_t - \bar{S}\right)^+, \quad f_T(T_t) = \left(\bar{T} - T_t\right)^+, \tag{23}$$

with $\bar{S}$ and $\bar{T}$ being predetermined thresholds, and $(x)^+ = \max(x, 0)$.

In such cases, the option price becomes

$$\mathcal{Q}(t_1, t_2) = \mathbb{E}^{\mathbb{Q}}\left[\sum_{t=t_1}^{t_2} \left(S_t - \bar{S}\right)^+ \left(\bar{T} - T_t\right)^+ \,\middle|\, \mathcal{F}_{t_0}\right]. \tag{24}$$

Since the temperature is not tradable and cannot be perfectly hedged, we may need to adjust our pricing approach. One method is to work under the real-world measure $\mathbb{P}$ for the temperature component and apply a risk adjustment to account for the market price of temperature risk, if any.

3.5. **Pricing Under the Real-World Measure.** Assuming that the time horizon is relatively short and the appropriate discount rate is uncertain, we can simplify the pricing by assuming a unit discount factor (i.e., $D(t_0, t_2) = 1$). The option price then becomes

$$\mathcal{Q}(t_1, t_2) = \mathbb{E}^{\mathbb{P}}\left[\sum_{t=t_1}^{t_2} \left(S_t - \bar{S}\right)^+ \left(\bar{T} - T_t\right)^+ \,\middle|\, \mathcal{F}_{t_0}\right]. \tag{25}$$

In this framework, we use the real-world measure $\mathbb{P}$ for both processes, acknowledging that the result may need to be adjusted for risk preferences or calibrated to market prices if available.

3.6. **Analytical Pricing of Energy Quanto Options.** In this subsection, we derive an analytical formula for pricing energy quanto options within the coupled CAR(p) model framework. These options have payoffs that depend on both the electricity spot price and the temperature, making the joint modeling of these variables essential for accurate valuation.

Consider an energy quanto option with payoff at time t defined as

$$\text{Payoff} = \sum_{t=t_1}^{t_2} \left(S_t - \bar{S}\right)^+ \left(\bar{T} - T_t\right)^+, \tag{26}$$

where $(x)^+ = \max(x, 0)$, $\bar{S}$ and $\bar{T}$ are predetermined thresholds, and S_t and T_t are the electricity spot price and temperature at time t, respectively.

Our goal is to compute the price of this option, given by

$$\mathcal{Q}(t_1, t_2) = \mathbb{E}\left[\sum_{t=t_1}^{t_2} \left(S_t - \bar{S}\right)^+ \left(\bar{T} - T_t\right)^+ \middle| \mathcal{F}_{t_0}\right], \tag{27}$$

where $\mathcal{F}_{t_0}$ represents the information available at the initial time t_0.

Theorem 3.1 (Quanto pricing formula)**.**

$$\begin{aligned}\mathcal{Q}(t_1, t_2) = \sum_{t=t_1}^{t_2} \Bigg(& \mathbb{E}_{\lambda=0}[(S_t - \bar{S})^+ | \mathcal{F}_{t_0}] \times \mathbf{e_1}' \Bigg[\boldsymbol{\mu}\Phi(-\Sigma^{-1/2}\boldsymbol{\mu}) \\ & + \frac{\Sigma^{1/2}\mathbf{e}}{\sqrt{(2\pi)^p}} \exp\left(-\frac{1}{2}(\Sigma^{-1/2}\boldsymbol{\mu})^\intercal(\Sigma^{-1/2}\boldsymbol{\mu})\right)\Bigg] \\ & + \left(\mathbb{E}_{\lambda=0}\left[(S_{t+\Delta} - \bar{S})^+ | \mathcal{F}_{t_0}\right] + \bar{S}\mathbb{P}(S_{t+\Delta} \geq \bar{S} | \mathcal{F}_{t_0})\right) \\ & \times \mathbf{e}_1' K_{XT}(\Delta) K_T^{-1}(\Delta)\Sigma^{1/2}\mathbf{e}\,\mathrm{Tr}\,(\Sigma^{1/2})\Phi(-\Sigma^{-1/2}\boldsymbol{\mu})\lambda\Bigg) + o(\lambda)\end{aligned}$$

Where $\boldsymbol{\mu} = \bar{T}\boldsymbol{e}_1 - e^{\Delta A_T}\boldsymbol{Z}_t^T$ *and* $\Sigma = \sigma_T^2 e^{\Delta A_T}\left(\hat{\mathcal{D}}_T^{-1}\left(\exp(\hat{\mathcal{D}}_T\Delta) - \mathbf{I}\right)\boldsymbol{e}_p\boldsymbol{e}_p'\right)e^{\Delta A_T^\intercal}$. Φ *is the cumulative distribution function of the multivariate normal distribution* $\mathcal{N}(\boldsymbol{0}, \mathrm{Id})$ *and* $\mathbf{e} := \begin{bmatrix}1 & 1 & \cdots & 1\end{bmatrix}^\intercal$.

Proof. [S: I have to revise the proof still. But look at page 27 for a way to get away of computing the multiple integral. Check if this holds.]

We aim to compute the first-order Taylor expansion of the expectation

$$\mathbb{E}\left[(S_{t+\Delta} - \bar{S})^+(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t\right] \tag{28}$$

with respect to λ at $\lambda = 0$.

Using the Taylor expansion, we have

$$\begin{aligned}\mathbb{E}\left[(S_{t+\Delta} - \bar{S})^+(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t\right] &= \mathbb{E}_{\lambda=0}\left[(S_{t+\Delta} - \bar{S})^+(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t\right] \\ &\quad + \lambda\frac{\partial}{\partial\lambda}\mathbb{E}\left[(S_{t+\Delta} - \bar{S})^+(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t\right]\Bigg|_{\lambda=0} + o(\lambda).\end{aligned} \tag{29}$$

When $\lambda = 0$, the spot price $S_{t+\Delta}$ and temperature $T_{t+\Delta}$ are independent, so

$$\mathbb{E}_{\lambda=0}\left[(S_{t+\Delta} - \bar{S})^+(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t\right] = \mathbb{E}_{\lambda=0}\left[(S_{t+\Delta} - \bar{S})^+ \middle| \mathcal{F}_t\right] \cdot \mathbb{E}\left[(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t\right]. \tag{30}$$

Step 1: Compute $\mathbb{E}\left[(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t\right]$

Recall that the temperature process is given by

$$T_{t+\Delta} = \mathbf{b}_T^\intercal \mathbf{Z}_{t+\Delta}^T,$$

where $\mathbf{Z}_{t+\Delta}^T$ is the state vector of the temperature process.

Define

$$\mathbf{X} = \bar{T}\mathbf{e}_1 - \mathbf{Z}_{t+\Delta}^T,$$

so that

$$\bar{T} - T_{t+\Delta} = \mathbf{b}_T^\intercal \mathbf{X}.$$

Under $\mathcal{F}_t$, the conditional distribution of $\mathbf{X}$ is multivariate normal:

$$\mathbf{X}\big|\mathcal{F}_t \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma),$$

where

$$\boldsymbol{\mu} = \bar{T}\mathbf{e}_1 - e^{\Delta A_T}\mathbf{Z}_t^T, \tag{31}$$

$$\Sigma = \sigma_T^2 e^{\Delta A_T}\left(\int_0^\Delta e^{-sA_T}\mathbf{e}_p\mathbf{e}_p^\intercal e^{-sA_T^\intercal}\,\mathrm{d}s\right)e^{\Delta A_T^\intercal}. \tag{32}$$

Let $Y = \mathbf{b}_T^\intercal\mathbf{X}$. Then Y is normally distributed with mean

$$\mu_Y = \mathbf{b}_T^\intercal\boldsymbol{\mu}$$

and variance

$$\sigma_Y^2 = \mathbf{b}_T^\intercal\Sigma\mathbf{b}_T.$$

Therefore,

$$\mathbb{E}\left[(\bar{T} - T_{t+\Delta})^+\middle|\mathcal{F}_t\right] = \mathbb{E}\left[(Y)^+\middle|\mathcal{F}_t\right] = \mu_Y\Phi\left(\frac{\mu_Y}{\sigma_Y}\right) + \sigma_Y\phi\left(\frac{\mu_Y}{\sigma_Y}\right), \tag{33}$$

where Φ and ϕ are the cumulative distribution function and probability density function of the standard normal distribution, respectively.

Step 2: Compute the Derivative of the Expectation with Respect to λ

The covariance between $S_{t+\Delta}$ and $T_{t+\Delta}$, to first order in λ, is given by

$$\mathrm{Cov}\left(S_{t+\Delta}, T_{t+\Delta}\middle|\mathcal{F}_t\right) = \lambda\cdot\text{Covariance Term} + o(\lambda).$$

Since $S_{t+\Delta}$ depends on W_t^T through the term $\lambda\sigma_T\mathbf{e}_p\,\mathrm{d}W_t^T$, the covariance can be computed using Itô's isometry.

Define the residuals:

$$\begin{cases}\mathbf{X}_T = \mathbf{Z}_{t+\Delta}^T - e^{\Delta A_T}\mathbf{Z}_t^T = \sigma_T e^{\Delta A_T}\displaystyle\int_t^{t+\Delta} e^{-(u-t)A_T}\mathbf{e}_p\,\mathrm{d}W_u^T,\\ \mathbf{X}_S = \mathbf{Z}_{t+\Delta}^X - e^{\Delta A_X}\mathbf{Z}_t^X = \lambda\sigma_T e^{\Delta A_X}\displaystyle\int_t^{t+\Delta} e^{-(u-t)A_X}\mathbf{e}_p\,\mathrm{d}W_u^T + \text{independent terms.}\end{cases}$$

The covariance between $\mathbf{X}_T$ and $\mathbf{X}_S$ is

$$\mathrm{Cov}\left(\mathbf{X}_T, \mathbf{X}_S\middle|\mathcal{F}_t\right) = \lambda\sigma_T^2 e^{\Delta A_T}\left(\int_0^\Delta e^{-sA_T}\mathbf{e}_p\mathbf{e}_p^\intercal e^{-sA_X^\intercal}\,\mathrm{d}s\right)e^{\Delta A_X^\intercal}.$$

Therefore, the covariance between $T_{t+\Delta}$ and $S_{t+\Delta}$ is

$$\mathrm{Cov}\left(T_{t+\Delta}, S_{t+\Delta}\middle|\mathcal{F}_t\right) = \lambda\sigma_T^2\mathbf{b}_T^\intercal e^{\Delta A_T}\left(\int_0^\Delta e^{-sA_T}\mathbf{e}_p\mathbf{e}_p^\intercal e^{-sA_X^\intercal}\,\mathrm{d}s\right)e^{\Delta A_X^\intercal}\mathbf{b}_X.$$

Step 3: Compute the Covariance of the Payoffs

Using the properties of jointly normally distributed variables and the fact that the payoffs are functions of $(S_{t+\Delta} - \bar{S})^+$ and $(\bar{T} - T_{t+\Delta})^+$, we can compute the covariance term.

Let $X_S = S_{t+\Delta}$ and $X_T = T_{t+\Delta}$.

Define the standardized variables:

$$\begin{cases}Z_S = \dfrac{X_S - \mu_S}{\sigma_S},\\ Z_T = \dfrac{X_T - \mu_T}{\sigma_T},\end{cases}$$

where μ_S, μ_T, σ_S, and σ_T are the means and standard deviations of $S_{t+\Delta}$ and $T_{t+\Delta}$, respectively.

The correlation coefficient is

$$\rho = \frac{\mathrm{Cov}(X_S, X_T)}{\sigma_S \sigma_T} = \lambda \cdot \rho_1,$$

where ρ_1 is the derivative of the correlation coefficient at $\lambda = 0$.

The covariance of the payoffs is then given by

$$\mathrm{Cov}\left((S_{t+\Delta} - \bar{S})^+, (\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t \right) = \sigma_S \sigma_T \rho \cdot \Theta,$$

where Θ is a function involving the bivariate normal cumulative distribution function.

To first order in λ, we can write

$$\mathrm{Cov}\left((S_{t+\Delta} - \bar{S})^+, (\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t \right) = \lambda \cdot \text{Covariance Term} + o(\lambda).$$

Step 1: Compute $\mathbb{E}\left[(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t \right]$

We have

$$\mathbb{E}\left[(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t \right] = \mathbb{E}\left[\left(\bar{T} - \mathbf{b}_T^\intercal \mathbf{Z}_{t+\Delta}^T\right)^+ \middle| \mathcal{F}_t \right].$$

Let us define

$$\mathbf{X} = \bar{T}\mathbf{e}_1 - \mathbf{Z}_{t+\Delta}^T,$$

so that

$$\bar{T} - T_{t+\Delta} = \mathbf{b}_T^\intercal \mathbf{X}.$$

Under $\mathcal{F}_t$, the conditional distribution of $\mathbf{X}$ is multivariate normal:

$$\mathbf{X} \big| \mathcal{F}_t \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma),$$

where

$$\boldsymbol{\mu} = \bar{T}\mathbf{e}_1 - e^{\Delta A_T} \mathbf{Z}_t^T, \tag{34}$$

$$\Sigma = \sigma_T^2 e^{\Delta A_T} \left(\hat{\mathcal{D}}_T^{-1} \left(e^{\hat{\mathcal{D}}_T \Delta} - \mathbf{I} \right) \mathbf{e}_p \mathbf{e}_p^\intercal \right) e^{\Delta A_T^\intercal}. \tag{35}$$

We can write

$$\mathbf{X} = \boldsymbol{\mu} + \Sigma^{1/2} \mathbf{Z},$$

where $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ under $\mathcal{F}_t$.

To compute $\mathbb{E}\left[(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t \right]$, we consider the expectation of a function of a linear combination of Gaussian variables. Specifically,

$$\mathbb{E}\left[(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t \right] = \mathbb{E}\left[(\mathbf{b}_T^\intercal \mathbf{X})^+ \middle| \mathcal{F}_t \right] \tag{36}$$

$$= \mathbb{E}\left[\left(\mathbf{b}_T^\intercal (\boldsymbol{\mu} + \Sigma^{1/2} \mathbf{Z}) \right)^+ \middle| \mathcal{F}_t \right]. \tag{37}$$

Let $Y = \mathbf{b}_T^\intercal (\boldsymbol{\mu} + \Sigma^{1/2} \mathbf{Z})$, which is a scalar normally distributed random variable with mean $\mu_Y = \mathbf{b}_T^\intercal \boldsymbol{\mu}$ and variance $\sigma_Y^2 = \mathbf{b}_T^\intercal \Sigma \mathbf{b}_T$.

Using the standard result for the expected positive part of a normal variable, we have

$$\mathbb{E}\left[(\bar{T} - T_{t+\Delta})^+ \middle| \mathcal{F}_t \right] = \mathbb{E}\left[Y^+ \middle| \mathcal{F}_t \right] \tag{38}$$

$$= \mu_Y \Phi\left(\frac{\mu_Y}{\sigma_Y} \right) + \sigma_Y \phi\left(\frac{\mu_Y}{\sigma_Y} \right), \tag{39}$$

where Φ and ϕ are the cumulative distribution function and probability density function of the standard normal distribution, respectively.

Step 2: Compute the Derivative of the Expectation with Respect to λ at $\lambda = 0$

We are interested in

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbb{E}\left[(S_{t+\Delta}-\bar{S})^+(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]\Bigg|_{\lambda=0}. \tag{40}$$

Using the product rule and the fact that at $\lambda = 0$, $S_{t+\Delta}$ and $T_{t+\Delta}$ are independent, we have

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbb{E}\left[(S_{t+\Delta}-\bar{S})^+(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]\Bigg|_{\lambda=0} = \mathbb{E}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda}(S_{t+\Delta}-\bar{S})^+\Big|_{\lambda=0}\cdot(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]. \tag{41}$$

Remark 3.2. *Since $T_{t+\Delta}$ does not depend on λ, its distribution remains the same when taking the derivative at $\lambda = 0$.*

We proceed to compute the derivative of $(S_{t+\Delta}-\bar{S})^+$ with respect to λ at $\lambda = 0$.

Recall that $S_{t+\Delta} = e^{X_{t+\Delta}}$, where $X_{t+\Delta} = \mathbf{b}_X^\intercal \mathbf{Z}_{t+\Delta}^X$.

At $\lambda = 0$, the process $\mathbf{Z}_{t+\Delta}^X$ does not depend on W_u^T (since λ multiplies the term involving W_u^T). Therefore, the derivative is

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\Big|_{\lambda=0} = \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\mathbf{b}_X^\intercal \mathbf{Z}_{t+\Delta}^X\right)\Big|_{\lambda=0} \tag{42}$$

$$= \mathbf{b}_X^\intercal\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbf{Z}_{t+\Delta}^X\Big|_{\lambda=0}\right). \tag{43}$$

From the dynamics of $\mathbf{Z}_{t+\Delta}^X$, we have

$$\mathbf{Z}_{t+\Delta}^X = e^{\Delta A_X}\mathbf{Z}_t^X + \lambda\sigma_T e^{\Delta A_X}\int_t^{t+\Delta} e^{-(u-t)A_X}\mathbf{e}_p\,\mathrm{d}W_u^T + \text{terms independent of } \lambda. \tag{44}$$

Thus,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbf{Z}_{t+\Delta}^X\Big|_{\lambda=0} = \sigma_T e^{\Delta A_X}\int_t^{t+\Delta} e^{-(u-t)A_X}\mathbf{e}_p\,\mathrm{d}W_u^T. \tag{45}$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\Big|_{\lambda=0} = \mathbf{b}_X^\intercal\left(\sigma_T e^{\Delta A_X}\int_t^{t+\Delta} e^{-(u-t)A_X}\mathbf{e}_p\,\mathrm{d}W_u^T\right). \tag{46}$$

Using the chain rule for the positive part function, we have

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}(S_{t+\Delta}-\bar{S})^+\Big|_{\lambda=0} = \mathbb{1}_{S_{t+\Delta}>\bar{S}}e^{X_{t+\Delta}}\,\frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\Big|_{\lambda=0}. \tag{47}$$

Computing the Derivative of the Expected Payoff

Now, the derivative becomes

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbb{E}\left[(S_{t+\Delta}-\bar{S})^+(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]\Bigg|_{\lambda=0} = \mathbb{E}_{\lambda=0}\left[\mathbb{1}_{S_{t+\Delta}>\bar{S}}e^{X_{t+\Delta}}\,\frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\Big|_{\lambda=0}(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]. \tag{48}$$

Since at $\lambda = 0$, $S_{t+\Delta}$ and $T_{t+\Delta}$ are independent, we can write

$$\mathbb{E}_{\lambda=0}\left[\mathbb{1}_{S_{t+\Delta}>\bar{S}}e^{X_{t+\Delta}}\middle|\mathcal{F}_t\right] = \mathbb{E}_{\lambda=0}\left[(S_{t+\Delta}-\bar{S})^+\middle|\mathcal{F}_t\right] + \bar{S}\mathbb{P}_{\lambda=0}\left(S_{t+\Delta}>\bar{S}\middle|\mathcal{F}_t\right). \tag{49}$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbb{E}\left[(S_{t+\Delta}-\bar{S})^+(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]\Big|_{\lambda=0} = \left(\mathbb{E}_{\lambda=0}\left[(S_{t+\Delta}-\bar{S})^+\middle|\mathcal{F}_t\right]+\bar{S}\mathbb{P}_{\lambda=0}\left(S_{t+\Delta}>\bar{S}\middle|\mathcal{F}_t\right)\right) \times \mathbb{E}_{\lambda=0}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\Big|_{\lambda=0}(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]. \tag{50}$$

Computing the Second Expectation

We need to compute

$$\mathbb{E}_{\lambda=0}\left[\frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\Big|_{\lambda=0}(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right] = \sigma_T\mathbf{b}_X^\intercal e^{\Delta A_X}\int_t^{t+\Delta}e^{-(u-t)A_X}\mathbf{e}_p\,\mathrm{d}u\cdot\mathbb{E}_{\lambda=0}\left[(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right], \tag{51}$$

but this is incorrect because the integral involves $\mathrm{d}W_u^T$, and $(\bar{T}-T_{t+\Delta})^+$ is a function of W_u^T.

We need to consider the joint distribution of

$$\left(\frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\Big|_{\lambda=0},\,\bar{T}-T_{t+\Delta}\right).$$

Using the Joint Normality

At $\lambda=0$, both $\frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\big|_{\lambda=0}$ and $\bar{T}-T_{t+\Delta}$ are linear functions of the same Brownian motion W_u^T. Therefore, their joint distribution is bivariate normal.

Let us define

$$U = \frac{\mathrm{d}}{\mathrm{d}\lambda}X_{t+\Delta}\Big|_{\lambda=0} = \sigma_T\mathbf{b}_X^\intercal e^{\Delta A_X}\int_t^{t+\Delta}e^{-(u-t)A_X}\mathbf{e}_p\,\mathrm{d}W_u^T, \tag{52}$$

$$V = \bar{T}-T_{t+\Delta} = \mathbf{b}_T^\intercal\left(\bar{T}\mathbf{e}_1-\mathbf{Z}_{t+\Delta}^T\right). \tag{53}$$

Both U and V are normally distributed with zero means and a certain covariance.

We can compute

$$\mathrm{Cov}(U,V) = \sigma_T^2\mathbf{b}_X^\intercal e^{\Delta A_X}\left(\int_t^{t+\Delta}e^{-(u-t)A_X}\mathbf{e}_p\mathbf{e}_p^\intercal e^{-(u-t)A_T^\intercal}\,\mathrm{d}u\right)e^{\Delta A_T^\intercal}\mathbf{b}_T. \tag{54}$$

Define the covariance matrix

$$K_{XT}(\Delta) = \sigma_T^2 e^{\Delta A_X}\hat{\mathcal{D}}_{XT}^{-1}\left(e^{\hat{\mathcal{D}}_{XT}\Delta}-\mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\intercal e^{\Delta A_T^\intercal}.$$

Therefore,

$$\mathrm{Cov}(U,V) = \mathbf{b}_X^\intercal K_{XT}(\Delta)\mathbf{b}_T.$$

Similarly, compute the variances

$$\mathrm{Var}(U) = \sigma_T^2\mathbf{b}_X^\intercal K_X(\Delta)\mathbf{b}_X,\quad \mathrm{Var}(V) = \mathbf{b}_T^\intercal\Sigma\mathbf{b}_T.$$

Computing the Expectation

Since U and V are jointly normal, we can write

$$\mathbb{E}\left[U\cdot(V)^+\middle|\mathcal{F}_t\right] = \mathrm{Cov}(U,V)\cdot\phi(0) = \mathrm{Cov}(U,V)\cdot\frac{1}{\sqrt{2\pi}\sigma_V}, \tag{55}$$

assuming that V has zero mean (we may need to adjust if V has non-zero mean).

However, since V has mean $\mu_V=\mathbf{b}_T^\intercal\boldsymbol{\mu}$, we need to consider the more general expression.

The expected value is

$$\mathbb{E}\left[U\cdot(V)^+\middle|\mathcal{F}_t\right] = \mathrm{Cov}(U,V)\cdot\phi\left(\frac{\mu_V}{\sigma_V}\right)+\mu_U\mathbb{E}\left[(V)^+\middle|\mathcal{F}_t\right], \tag{56}$$

where $\mu_U=0$.

Since $\mu_U = 0$, the second term vanishes.

Thus, the derivative of the expectation is

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathbb{E}\left[(S_{t+\Delta}-\bar{S})^+(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]\Bigg|_{\lambda=0} = \left(\mathbb{E}_{\lambda=0}\left[(S_{t+\Delta}-\bar{S})^+\middle|\mathcal{F}_t\right] + \bar{S}\mathbb{P}_{\lambda=0}\left(S_{t+\Delta}>\bar{S}\middle|\mathcal{F}_t\right)\right) \times \frac{\mathrm{Cov}(U,V)}{\sigma_V}\phi\left(\frac{\mu_V}{\sigma_V}\right). \tag{57}$$

Final Expression

Putting everything together, we have

$$\begin{aligned}\mathcal{Q}(t_1,t_2) = \sum_{t=t_1}^{t_2}\Bigg\{&\mathbb{E}_{\lambda=0}\left[(S_t-\bar{S})^+\middle|\mathcal{F}_{t_0}\right]\cdot\mathbb{E}\left[(\bar{T}-T_t)^+\middle|\mathcal{F}_{t_0}\right]\\ &+\lambda\left(\mathbb{E}_{\lambda=0}\left[(S_t-\bar{S})^+\middle|\mathcal{F}_{t_0}\right]+\bar{S}\mathbb{P}_{\lambda=0}\left(S_t>\bar{S}\middle|\mathcal{F}_{t_0}\right)\right)\\ &\times\frac{\mathrm{Cov}(U,V)}{\sigma_V}\phi\left(\frac{\mu_V}{\sigma_V}\right)\Bigg\}+o(\lambda).\end{aligned} \tag{58}$$

□

Now, we aim to compute

$$\mathbb{E}\left[\mathbf{b}_X^\intercal\sigma_T\mathbf{X}\cdot(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right], \tag{59}$$

where $T_{t+\Delta} = \mathbf{b}_T^\intercal\mathbf{Z}_{t+\Delta}^T$.

Using the law of total expectation and the properties of conditional expectations, we have

$$\begin{aligned}&\mathbb{E}\left[\mathbf{b}_X^\intercal\sigma_T\mathbf{X}\cdot(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]\\ &=\mathbb{E}\left[\mathbb{E}\left[\mathbf{b}_X^\intercal\sigma_T\mathbf{X}\cdot(\bar{T}-T_{t+\Delta})^+\middle|\mathbf{Y}\right]\middle|\mathcal{F}_t\right]\\ &=\mathbb{E}\left[\mathbf{b}_X^\intercal\sigma_T\mathbb{E}\left[\mathbf{X}\middle|\mathbf{Y}\right]\cdot(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right]\\ &=\mathbf{b}_X^\intercal\sigma_T K_{XT}(\Delta)K_T^{-1}(\Delta)\mathbb{E}\left[\mathbf{Y}\cdot(\bar{T}-T_{t+\Delta})^+\middle|\mathcal{F}_t\right].\end{aligned} \tag{60}$$

Note that $\mathbf{Y}$ is related to $T_{t+\Delta}$ via

$$T_{t+\Delta} = \mathbf{b}_T^\intercal\left(e^{\Delta A_T}\mathbf{Z}_t^T+\sigma_T\mathbf{Y}\right). \tag{61}$$

Define

$$\boldsymbol{\mu} = \bar{T}-\mathbf{b}_T^\intercal e^{\Delta A_T}\mathbf{Z}_t^T, \tag{62}$$

$$\Sigma_T = \sigma_T^2\mathbf{b}_T^\intercal e^{\Delta A_T}\hat{\mathcal{D}}_T^{-1}\left(e^{\hat{\mathcal{D}}_T\Delta}-\mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\intercal e^{\Delta A_T^\intercal}\mathbf{b}_T. \tag{63}$$

Then,

$$\bar{T}-T_{t+\Delta} = \boldsymbol{\mu}-\sigma_T\mathbf{b}_T^\intercal\mathbf{Y}. \tag{64}$$

Let $V = \bar{T}-T_{t+\Delta} = \boldsymbol{\mu}-\sigma_T\mathbf{b}_T^\intercal\mathbf{Y}$ and $U = \mathbf{b}_Y^\intercal\mathbf{Y}$, where

$$\mathbf{b}_Y = \sigma_T K_T^{-1}(\Delta)^\intercal K_{XT}(\Delta)^\intercal\mathbf{b}_X. \tag{65}$$

Our expectation becomes

$$\mathbb{E}\left[U \cdot V^{+} \middle| \mathcal{F}_t\right]. \tag{66}$$

Since U and V are linear functions of the same Gaussian vector $\mathbf{Y}$, they are jointly normally distributed. We can compute their covariance and variances:

$$\mathrm{Var}(U) = \mathbf{b}_Y^{\intercal} K_T(\Delta) \mathbf{b}_Y, \tag{67}$$

$$\mathrm{Var}(V) = \sigma_T^2 \mathbf{b}_T^{\intercal} K_T(\Delta) \mathbf{b}_T, \tag{68}$$

$$\mathrm{Cov}(U, V) = -\sigma_T \mathbf{b}_Y^{\intercal} K_T(\Delta) \mathbf{b}_T. \tag{69}$$

The negative sign in the covariance arises from the minus sign in the expression for V.

We can standardize U and V:

$$\tilde{U} = \frac{U}{\sigma_U}, \quad \sigma_U = \sqrt{\mathrm{Var}(U)}, \tag{70}$$

$$\tilde{V} = \frac{V - \mu_V}{\sigma_V}, \quad \mu_V = \boldsymbol{\mu}, \quad \sigma_V = \sqrt{\mathrm{Var}(V)}. \tag{71}$$

The correlation coefficient is

$$\rho = \frac{\mathrm{Cov}(U, V)}{\sigma_U \sigma_V}. \tag{72}$$

Therefore, the expectation can be expressed as

$$\mathbb{E}\left[U \cdot V^{+}\right] = \sigma_U \sigma_V \left(\rho \phi\left(\frac{\mu_V}{\sigma_V}\right) + \left(\frac{\mu_V}{\sigma_V}\rho^2 + \rho\right) \Phi\left(-\frac{\mu_V}{\sigma_V}\right)\right), \tag{73}$$

where ϕ and Φ are the standard normal PDF and CDF, respectively.

Given that $\mu_U = \mathbb{E}[U] = 0$, the expression simplifies, and we obtain

$$\mathbb{E}\left[U \cdot V^{+}\right] = -\,\mathrm{Cov}(U, V) \phi\left(\frac{\mu_V}{\sigma_V}\right). \tag{74}$$

Substituting back, we have

$$\mathbb{E}\left[\mathbf{b}_X^{\intercal} \sigma_T \mathbf{X} \cdot (\bar{T} - T_{t+\Delta})^{+} \middle| \mathcal{F}_t\right] = -\,\mathrm{Cov}(U, V) \phi\left(\frac{\mu_V}{\sigma_V}\right). \tag{75}$$

Therefore, the final expression is

$$\mathbb{E}\left[\mathbf{b}_X^{\intercal} \sigma_T \mathbf{X} \cdot (\bar{T} - T_{t+\Delta})^{+} \middle| \mathcal{F}_t\right] = \sigma_T^2 \mathbf{b}_X^{\intercal} K_{XT}(\Delta) K_T^{-1}(\Delta) \mathbf{b}_T \phi\left(\frac{\boldsymbol{\mu}}{\sigma_V}\right). \tag{76}$$

4. Hedging of Energy Quanto Options in the coupled CARMA Model

Effective hedging requires calculating the sensitivities of the option price to various risk factors, commonly known as the Greeks.

4.0.1. *Delta Hedging.* The Delta of the option with respect to the electricity price X_t is given by:

$$\Delta_X = \frac{\partial V_0}{\partial X_0}. \tag{77}$$

Similarly, the Delta with respect to the temperature T_t is:

$$\Delta_T = \frac{\partial V_0}{\partial T_0}. \tag{78}$$

These can be computed numerically using finite difference methods or by differentiating the pricing formula if analytical expressions are available.

4.0.2. *Gamma and Vega.* Higher-order Greeks such as Gamma and Vega provide insights into the curvature of the option price with respect to underlying variables and volatility, respectively:

$$\Gamma_X = \frac{\partial^2 V_0}{\partial X_0^2}, \quad \text{Vega} = \frac{\partial V_0}{\partial \sigma_X}. \tag{79}$$

4.0.3. *Hedging Portfolio Construction.* By combining positions in the underlying assets and possibly other derivatives, we can construct a hedging portfolio that minimizes the risk associated with movements in X_t and T_t.

Appendix A. Linear State-Space Models and CARMA Processes

In this section, we consider linear state-space models where the state and output processes take values in $\mathbb{R}$. Let $p \in \mathbb{N}$, and let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions. We assume that this space is rich enough to support a real-valued two-sided Lévy process $L = (L_t)_{t\in\mathbb{R}}$ with finite moments as needed.

Definition A.1. *Let the tuple (A, B, C, L) consist of:*

- *A* state-transition matrix $A \in \mathbb{R}^{p\times p}$, *defined by*

$$A = \begin{pmatrix} -a_1 & -a_2 & \cdots & -a_{p-1} & -a_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix},$$

  where $a_i \in \mathbb{R}$ are scalar coefficients.
- *An* input vector $B \in \mathbb{R}^p$, *defined by*

$$B = \begin{pmatrix} b_0 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

  where $b_0 \in \mathbb{R}$ is a scalar coefficient.

- *An* output vector $C \in \mathbb{R}^p$*, defined by*
$$C = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_p \end{pmatrix},$$
where $c_i \in \mathbb{R}$ *are scalar coefficients.*
- *A real-valued two-sided Lévy process* $L = (L_t)_{t\in\mathbb{R}}$.

A continuous-time linear state-space model *associated with the parameter set* (A, B, C, L) *consists of:*

- *A* state equation*:*
$$\mathrm{d}\mathbf{Z}_t = A\mathbf{Z}_t \,\mathrm{d}t + B \,\mathrm{d}L_t, \quad t \in \mathbb{R}, \tag{80}$$
where $\mathbf{Z}_t \in \mathbb{R}^p$ *is the state vector.*
- *An* output equation*:*
$$X_t = C^\top \mathbf{Z}_t, \quad t \in \mathbb{R}, \tag{81}$$
where $X_t \in \mathbb{R}$ *is the output process.*

We refer to $(\mathbf{Z}_t)_{t\in\mathbb{R}}$ *as the* state process *and* $(X_t)_{t\in\mathbb{R}}$ *as the* output process *of the continuous-time linear state-space model associated with* (A, B, C, L).

Remark A.2. *In this setup, the matrices* A*,* B*, and* C *are defined to ensure dimensional consistency. The coefficients* a_i*,* b_0*, and* c_i *are real numbers that determine the dynamics of the system. The state vector* $\mathbf{Z}_t$ *is an element of* $\mathbb{R}^p$*, representing the internal state of the system at time* t.

The state equation (80) can be written component-wise as
$$\mathrm{d}Z_t^{(1)} = \left(-a_1 Z_t^{(1)} - a_2 Z_t^{(2)} - \cdots - a_p Z_t^{(p)}\right) \mathrm{d}t + b_0 \,\mathrm{d}L_t,$$
$$\mathrm{d}Z_t^{(i)} = Z_t^{(i-1)} \,\mathrm{d}t, \quad \text{for } i = 2, \ldots, p.$$

The solution to the state equation is given by the variation-of-constants formula:
$$\mathbf{Z}_t = e^{A(t-s)}\mathbf{Z}_s + \int_s^t e^{A(t-u)} B \,\mathrm{d}L_u, \quad s < t \in \mathbb{R}. \tag{82}$$

Remark A.3. *The matrix exponential* e^{At} *is defined via the usual exponential series expansion for matrices. Since* A *is a finite-dimensional matrix, this exponential is well-defined and can be computed using standard methods.*

Definition A.4. *We call a continuous-time state-space model associated with* (A, B, C, L) stable *if all eigenvalues of* A *have negative real parts. This ensures that the homogeneous solution decays to zero over time. If, in addition,* L *has finite logarithmic moments (i.e.,* $\mathbb{E}[\log|L_1|] < \infty$*), then there exists a unique stationary solution to* (80) *given by*
$$\mathbf{Z}_t = \int_{-\infty}^t e^{A(t-s)} B \,\mathrm{d}L_s, \quad t \in \mathbb{R}. \tag{83}$$
In this case, we refer to $(\mathbf{Z}_t)_{t\in\mathbb{R}}$ *as the* stable state process *and* $(X_t)_{t\in\mathbb{R}}$ *as the* stable output process*. If* A *has eigenvalues with non-negative real parts, we call the state-space model* unstable*, and solutions may not converge to a stationary process without additional conditions.*

Remark A.5. *The condition on the eigenvalues of* A *ensures exponential stability of the deterministic part of the state equation. The existence of a stationary solution in the stable case follows from standard results in stochastic differential equations driven by Lévy processes (see [**? ?**]).*

Remark A.6. *When $p = 1$, our definition coincides with the standard continuous-time state-space models for scalar Ornstein-Uhlenbeck processes. In this case, the state equation simplifies to*

$$\mathrm{d}Z_t = -a_1 Z_t \, \mathrm{d}t + b_0 \, \mathrm{d}L_t,$$

and the output process is simply $X_t = c_1 Z_t$.

A.1. **CARMA Processes.** Continuous-time autoregressive moving average (CARMA) processes extend the concept of ARMA processes to continuous time. They can be represented using the state-space framework described above.

A CARMA process of order (p, q), with $p > q$, is defined as the output process $\{X_t\}_{t\in\mathbb{R}}$ of a state-space model where the state equation corresponds to a system whose characteristic polynomial matches that of the desired CARMA process.

The CARMA process satisfies a higher-order stochastic differential equation:

$$D^p X_t + a_1 D^{p-1} X_t + \cdots + a_p X_t = b_0 D^q L_t + b_1 D^{q-1} L_t + \cdots + b_q L_t, \tag{84}$$

where $D = \dfrac{\mathrm{d}}{\mathrm{d}t}$ denotes differentiation, a_i and b_j are real coefficients, and L_t is a real-valued Lévy process.

The correspondence between the state-space representation and the higher-order SDE allows us to utilize the powerful machinery of state-space models to analyze and simulate CARMA processes.

Remark A.7. *The use of higher-order CARMA processes enables modeling of time series with complex dependencies and autocorrelation structures, which are common in applications such as finance and engineering.*

Given a state process $\{\mathbf{Z}_t\}_{t\in\mathbb{R}}$, we now shift our focus to the output process $\{X_t\}_{t\in\mathbb{R}}$ defined by:

$$X_t = C^\top \mathbf{Z}_t, \quad t \in \mathbb{R}. \tag{85}$$

In the next proposition, we summarize some well-known and fundamental properties of the output process $\{X_t\}_{t\in\mathbb{R}}$.

Proposition A.8. *Let (A, B, C, L) be as in Definition A.1, and let ψ_L be the characteristic exponent of the Lévy process L. Then, the output process $\{X_t\}_{t\in\mathbb{R}}$ defined by equation (85) satisfies the following properties:*

(1) ***Variation of Constants Formula****: For any $s < t \in \mathbb{R}$,*

$$X_t = C^\top e^{A(t-s)} \mathbf{Z}_s + \int_s^t C^\top e^{A(t-u)} B \, \mathrm{d}L_u. \tag{86}$$

(2) ***Conditional Characteristic Function****: For every $\theta \in \mathbb{R}$ and $\mathcal{F}_s$, the conditional characteristic function of X_t given $\mathcal{F}_s$ is*

$$\mathbb{E}\left[e^{\mathrm{i}\theta X_t} \middle| \mathcal{F}_s \right] = \exp\left(\mathrm{i}\theta C^\top e^{A(t-s)} \mathbf{Z}_s + \int_s^t \psi_L \left(\theta C^\top e^{A(t-u)} B \right) \mathrm{d}u \right). \tag{87}$$

(3) ***Conditional Mean****: If L is integrable (i.e., $\mathbb{E}[|L_1|] < \infty$), then the conditional mean of X_t is finite and given by*

$$\mathbb{E}[X_t | \mathcal{F}_s] = C^\top e^{A(t-s)} \mathbf{Z}_s + \int_s^t C^\top e^{A(t-u)} B \, \mu_L \, \mathrm{d}u, \tag{88}$$

where $\mu_L = \mathbb{E}[L_1]$ is the mean of the Lévy process L.

(4) ***Stationary Case**: If L has finite logarithmic moments (i.e., $\mathbb{E}[\log|L_1|] < \infty$), and the output process $\{X_t\}_{t\in\mathbb{R}}$ is stable (i.e., all eigenvalues of A have negative real parts), then $\{X_t\}_{t\in\mathbb{R}}$ is stationary and given by*

$$X_t = \int_{-\infty}^{t} C^\top e^{A(t-u)} B \, \mathrm{d}L_u, \quad t \in \mathbb{R}. \tag{89}$$

Moreover, if L is integrable, then the mean of X_t is constant (independent of t) and given by

$$\mathbb{E}[X_t] = -C^\top A^{-1} B \, \mu_L, \quad t \in \mathbb{R}. \tag{90}$$

Proof. We provide brief explanations for each part.

(1) **Variation of Constants**: The expression for X_t follows directly from substituting the solution of the state equation for $\mathbf{Z}_t$ into $X_t = C^\top \mathbf{Z}_t$, and applying the variation-of-constants formula for $\mathbf{Z}_t$:

$$\mathbf{Z}_t = e^{A(t-s)}\mathbf{Z}_s + \int_s^t e^{A(t-u)} B \, \mathrm{d}L_u.$$

Thus,

$$X_t = C^\top \mathbf{Z}_t = C^\top e^{A(t-s)}\mathbf{Z}_s + \int_s^t C^\top e^{A(t-u)} B \, \mathrm{d}L_u.$$

(2) **Conditional Characteristic Function**: Using the independence of Lévy increments and linearity, we have

$$\begin{aligned}\mathbb{E}\left[e^{\mathrm{i}\theta X_t}\middle|\mathcal{F}_s\right] &= e^{\mathrm{i}\theta C^\top e^{A(t-s)}\mathbf{Z}_s} \cdot \mathbb{E}\left[e^{\mathrm{i}\theta \int_s^t C^\top e^{A(t-u)} B \, \mathrm{d}L_u}\middle|\mathcal{F}_s\right] \\ &= e^{\mathrm{i}\theta C^\top e^{A(t-s)}\mathbf{Z}_s} \cdot \exp\left(\int_s^t \psi_L\left(\theta C^\top e^{A(t-u)} B\right) \mathrm{d}u\right),\end{aligned}$$

where we have utilized the fact that the characteristic function of a stochastic integral with respect to L involves the Lévy exponent ψ_L applied to the integrand.

(3) **Conditional Mean**: If L is integrable, we compute the conditional mean by taking expectations inside the integral:

$$\begin{aligned}\mathbb{E}[X_t|\mathcal{F}_s] &= C^\top e^{A(t-s)}\mathbf{Z}_s + \int_s^t C^\top e^{A(t-u)} B \, \mathbb{E}[\mathrm{d}L_u] \\ &= C^\top e^{A(t-s)}\mathbf{Z}_s + \int_s^t C^\top e^{A(t-u)} B \, \mu_L \, \mathrm{d}u.\end{aligned}$$

(4) **Stationary Case**: For a stable system, the solution converges to the stationary solution:

$$X_t = \int_{-\infty}^{t} C^\top e^{A(t-u)} B \, \mathrm{d}L_u.$$

The mean of X_t is then

$$\begin{aligned}\mathbb{E}[X_t] &= \int_{-\infty}^{t} C^\top e^{A(t-u)} B \, \mu_L \, \mathrm{d}u \\ &= C^\top \left(\int_0^\infty e^{Av} \, \mathrm{d}v\right) B \, \mu_L \\ &= -C^\top A^{-1} B \, \mu_L,\end{aligned}$$

since the integral of e^{Av} from 0 to ∞ equals $-A^{-1}$ when A is stable.

□

Appendix B. Proof of Theorem 3.1

Lemma B.1. *Let X be a normally distributed random variable with mean μ and standard deviation σ. Then, the expected positive part of X is given by*

$$\mathbb{E}[X^+] = \mu\Phi\left(\frac{\mu}{\sigma}\right) + \sigma\phi\left(\frac{\mu}{\sigma}\right), \tag{91}$$

where Φ is the cumulative distribution function and ϕ is the probability density function of the standard normal distribution.

Proof. The expected positive part is

$$\mathbb{E}[X^+] = \int_0^\infty x \cdot \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \mathrm{d}x \tag{92}$$

$$= \mu\Phi\left(\frac{\mu}{\sigma}\right) + \sigma\phi\left(\frac{\mu}{\sigma}\right). \tag{93}$$

□

Lemma B.2. *Let X and Y be jointly normally distributed with means μ_X, μ_Y, standard deviations σ_X, σ_Y, and correlation coefficient ρ. Then, the covariance of the positive parts is given by*

$$\mathrm{Cov}\left(X^+, Y^+\right) = \sigma_X\sigma_Y\rho \cdot \Theta, \tag{94}$$

where Θ is a function depending on μ_X, μ_Y, σ_X, σ_Y, and ρ.

Proof. The covariance can be computed using the properties of the bivariate normal distribution and involves evaluating integrals of the form

$$\int_0^\infty \int_0^\infty (x-\mu_X)(y-\mu_Y) f_{X,Y}(x,y)\, \mathrm{d}x\, \mathrm{d}y,$$

where $f_{X,Y}(x,y)$ is the joint probability density function of X and Y. □

Remark B.3. *The function Θ can be expressed in terms of the cumulative distribution function of the bivariate normal distribution, which can be evaluated numerically.*

Proposition B.4. *The joint distribution of the random vectors*

$$\left(\int_t^{t+\Delta} e^{(t+\Delta-u)A_X}\mathbf{e}_p\, \mathrm{d}W_u^T, \int_t^{t+\Delta} e^{(t+\Delta-u)A_T}\mathbf{e}_p\, \mathrm{d}W_u^T\right)$$

is a zero-mean multivariate normal distribution with covariance matrix

$$\mathbf{K}(\Delta) = \begin{bmatrix} K_X(\Delta) & K_{XT}(\Delta) \\ K_{XT}^\intercal(\Delta) & K_T(\Delta) \end{bmatrix},$$

where

$$K_X(\Delta) = \sigma_T^2 e^{\Delta A_X}\hat{\mathcal{D}}_X^{-1}\left(e^{\hat{\mathcal{D}}_X\Delta} - \mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\intercal e^{\Delta A_X^\intercal}, \tag{95}$$

$$K_T(\Delta) = \sigma_T^2 e^{\Delta A_T}\hat{\mathcal{D}}_T^{-1}\left(e^{\hat{\mathcal{D}}_T\Delta} - \mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\intercal e^{\Delta A_T^\intercal}, \tag{96}$$

$$K_{XT}(\Delta) = \sigma_T^2 e^{\Delta A_X}\hat{\mathcal{D}}_{XT}^{-1}\left(e^{\hat{\mathcal{D}}_{XT}\Delta} - \mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\intercal e^{\Delta A_T^\intercal}. \tag{97}$$

The linear operators $\hat{\mathcal{D}}_X$, $\hat{\mathcal{D}}_T$, and $\hat{\mathcal{D}}_{XT}$ are defined on $\mathbb{M}_p$, the space of $p \times p$ real matrices, by

$$\hat{\mathcal{D}}_X(M) = A_X M + M A_X^\intercal, \tag{98}$$

$$\hat{\mathcal{D}}_T(M) = A_T M + M A_T^\intercal, \tag{99}$$

$$\hat{\mathcal{D}}_{XT}(M) = A_X M + M A_T^\intercal. \tag{100}$$

Proof. The covariance matrices are derived using Itô's isometry and properties of matrix exponentials. The integrals involve the same Brownian motion W_u^T, leading to non-zero covariance between the two integrals. □

Corollary B.5. *Conditionally on* $\mathbf{Y} = \int_t^{t+\Delta} e^{(t+\Delta-u)A_T} \boldsymbol{e}_p \, \mathrm{d}W_u^T$, *the random vector* $\mathbf{X} = \int_t^{t+\Delta} e^{(t+\Delta-u)A_X} \boldsymbol{e}_p \, \mathrm{d}W_u^T$ *follows a Gaussian distribution with mean*

$$\mathbb{E}\left[\mathbf{X}\middle|\mathbf{Y}\right] = K_{XT}(\Delta)K_T^{-1}(\Delta)\mathbf{Y}, \tag{101}$$

and covariance matrix

$$\mathrm{Var}\left(\mathbf{X}\middle|\mathbf{Y}\right) = K_X(\Delta) - K_{XT}(\Delta)K_T^{-1}(\Delta)K_{TX}(\Delta), \tag{102}$$

where

$$\begin{aligned} K_X(\Delta) &= \sigma_T^2 e^{\Delta A_X}\hat{\mathcal{D}}_X^{-1}\left(e^{\hat{\mathcal{D}}_X\Delta} - \mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\intercal e^{\Delta A_X^\intercal}, \\ K_T(\Delta) &= \sigma_T^2 e^{\Delta A_T}\hat{\mathcal{D}}_T^{-1}\left(e^{\hat{\mathcal{D}}_T\Delta} - \mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\intercal e^{\Delta A_T^\intercal}, \\ K_{XT}(\Delta) &= \sigma_T^2 e^{\Delta A_X}\hat{\mathcal{D}}_{XT}^{-1}\left(e^{\hat{\mathcal{D}}_{XT}\Delta} - \mathbf{I}\right)\mathbf{e}_p\mathbf{e}_p^\intercal e^{\Delta A_T^\intercal}. \end{aligned}$$

Proof. Since $\mathbf{X}$ and $\mathbf{Y}$ are jointly Gaussian with zero mean, their conditional distribution is Gaussian with the given mean and covariance, derived from the standard formulas for conditioning in multivariate normal distributions. □

Lemma B.6. *Let X and Y be jointly normally distributed random variables with $\mathbb{E}[X] = 0$, $\mathbb{E}[Y] = \mu_Y$, variances σ_X^2, σ_Y^2, and correlation coefficient ρ. Then,*

$$\mathbb{E}[X \cdot Y^+] = -\sigma_X\sigma_Y\rho\phi\left(\frac{\mu_Y}{\sigma_Y}\right). \tag{103}$$

Proof. Since X and Y are jointly normal, the expectation $\mathbb{E}[X \cdot Y^+]$ can be computed using the properties of the bivariate normal distribution. The derivation involves integrating over the joint density and leveraging the symmetry and properties of the normal distribution. □

Lemma B.7. *Let X and Y be jointly normally distributed random variables with $\mathbb{E}[X] = 0$, $\mathbb{E}[Y] = \mu_Y$, variances σ_X^2, σ_Y^2, and correlation coefficient ρ. Then,*

$$\mathbb{E}[XY^+] = \sigma_X\sigma_Y\rho\phi\left(\frac{\mu_Y}{\sigma_Y}\right) + \mu_Y\mathbb{E}[X\Phi\left(\frac{X\rho - \mu_Y}{\sigma_Y\sqrt{1-\rho^2}}\right)],$$

where ϕ is the standard normal probability density function and Φ is the cumulative distribution function.

We aim to compute the integral

$$\mathbf{I}_1 = \int_{\mathbf{x}\geq -A^{-1}\mathbf{b}} A\mathbf{x}\mathbf{e}_1' A\mathbf{x}\frac{1}{\sqrt{(2\pi)^p}}\exp\left(-\frac{1}{2}\mathbf{x}^\intercal\mathbf{x}\right)\mathrm{d}\mathbf{x}.$$

However, directly computing this integral is complex due to the multidimensional integration over a truncated domain. Instead, we can simplify the problem by leveraging properties of multivariate normal distributions and known results.

Let us consider the random vector

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b},$$

where $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Then, $\mathbf{Y} \sim \mathcal{N}(\mathbf{b}, AA^\intercal)$.

Our goal is to compute

$$\mathbb{E}\left[(A\mathbf{X})\mathbf{e}_1'\mathbf{Y}^+\right] = \mathbb{E}\left[(\mathbf{Y}-\mathbf{b})\mathbf{e}_1'\mathbf{Y}^+\right],$$

where $\mathbf{Y}^+$ denotes the vector obtained by taking the positive part of each component of $\mathbf{Y}$.

We focus on computing each component of the vector $\mathbb{E}\left[(\mathbf{Y}-\mathbf{b})\mathbf{e}_1'\mathbf{Y}^+\right]$. Specifically, for each $i = 1, \ldots, p$, we need to compute

$$\mathbb{E}\left[(Y_i - b_i)Y_1^+\right].$$

Since $Y_i - b_i$ and Y_1 are jointly normally distributed with mean zero and covariance $(AA^\intercal)_{i1}$, we can use properties of the bivariate normal distribution.

However, since $\mathbb{E}[X] = 0$ and X and Y are jointly normal, the term involving $\mathbb{E}[X\Phi(\cdot)]$ simplifies to zero due to symmetry. Therefore, we have

$$\mathbb{E}[XY^+] = \sigma_{XY}\phi\left(\frac{\mu_Y}{\sigma_Y}\right),$$

where $\sigma_{XY} = \sigma_X\sigma_Y\rho$.

Applying this to our case:

- $X = Y_i - b_i$, which has mean zero and variance $(AA^\intercal)_{ii}$.
- $Y = Y_1$, with mean $\mu_{Y_1} = b_1$ and variance $(AA^\intercal)_{11}$.
- The covariance between X and Y is $\sigma_{XY} = \mathrm{Cov}(Y_i - b_i, Y_1) = (AA^\intercal)_{i1}$.

Therefore,

$$\mathbb{E}\left[(Y_i - b_i)Y_1^+\right] = (AA^\intercal)_{i1}\phi\left(\frac{b_1}{\sqrt{(AA^\intercal)_{11}}}\right).$$

Collecting the components, we obtain

$$\mathbb{E}\left[(A\mathbf{X})\mathbf{e}_1'\mathbf{Y}^+\right] = A\mathbf{e}\cdot(AA^\intercal)_1\phi\left(\frac{b_1}{\sqrt{(AA^\intercal)_{11}}}\right),$$

where $(AA^\intercal)_1$ denotes the first column of $AA^\intercal$.

Final Expression.

Since $(AA^\intercal)_1$ is A times the first column of $A^\intercal$, and $A^\intercal$ has columns $(a_1, \ldots, a_p)$, where a_i are the rows of A, we have

$$(AA^\intercal)_1 = AA^\intercal\mathbf{e}_1 = Aa_1.$$

Therefore,

$$\mathbb{E}\left[A\mathbf{X}\mathbf{e}_1'\mathbf{Y}^+\right] = A\mathbf{e}\cdot Aa_1\phi\left(\frac{b_1}{\sqrt{(AA^\intercal)_{11}}}\right).$$

Simplifying further, since Aa_1 is a vector, and we are taking the dot product with $A\mathbf{e}$, the expression becomes

$$\mathbb{E}\left[A\mathbf{X}\mathbf{e}_1'\mathbf{Y}^+\right] = (A\mathbf{e})\,(Aa_1)^\intercal\,\phi\left(\frac{b_1}{\sqrt{(AA^\intercal)_{11}}}\right).$$

This expression can be further simplified depending on the specific structure of A.

References